

On the Erdős–Ulam problem for monochromatic union-closed families

By DEEP BHATTACHARJEE[†], PRIYABRATA MANDAL[‡], and USHASHI BHATTACHARYA

Abstract

We prove the two Erdős–Ulam conjectures on monochromatic union-closed families. For every fixed exponent, each two-colouring of the subsets of a large finite set contains a monochromatic union-closed family larger than that power of the size of the set, while suitable colourings keep every such family subexponential. The lower bound combines a supersaturated Hales–Jewett theorem with chains of disjoint blocks; for colourings by cardinality it becomes a statement about monochromatic Hilbert cubes. Both proofs are checked in Lean.

1. Introduction

Write $[n] = \{1, \dots, n\}$ and $2^{[n]}$ for the family of all subsets of $[n]$. A 2-colouring is a map $\chi : 2^{[n]} \rightarrow \{0, 1\}$, and a family $\mathcal{F} \subseteq 2^{[n]}$ is *monochromatic* if χ is constant on it. We call $\mathcal{F}$ *union-closed* if $A \cup B \in \mathcal{F}$ whenever $A, B \in \mathcal{F}$, and a *sublattice* if it is closed under both union and intersection. Erdős and Ulam asked how large a monochromatic family of either kind every colouring is forced to contain. Accordingly let

$$f(n) = \min_{\chi} \max\{|\mathcal{L}| : \mathcal{L} \subseteq 2^{[n]} \text{ is a monochromatic sublattice}\},$$

$$F(n) = \min_{\chi} \max\{|\mathcal{F}| : \mathcal{F} \subseteq 2^{[n]} \text{ is a monochromatic union-closed family}\},$$

the minima being taken over all 2-colourings of $2^{[n]}$. A single set is a sublattice, and every sublattice is union-closed, so $1 \leq f(n) \leq F(n)$. A maximal chain

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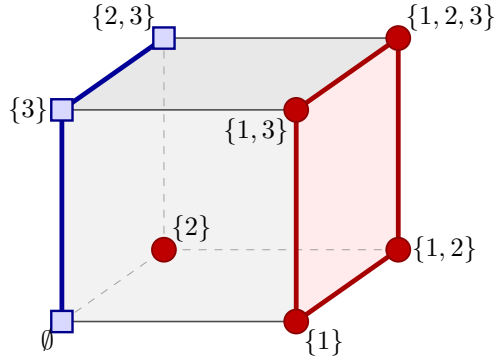


Figure 1. A 2-colouring of the subsets of $\{1, 2, 3\}$, drawn on the cube $2^{[3]}$. The five red sets (discs) form a union-closed family which is not a sublattice, since $\{1\} \cap \{2\} = \emptyset$ is blue. The three blue sets (squares) form a chain.

$\emptyset \subset \{1\} \subset \{1, 2\} \subset \cdots \subset [n]$ has $n + 1$ members, at least $\lceil (n + 1)/2 \rceil$ of them receive the same colour, and every subchain is a sublattice. Hence

$$F(n) \geq f(n) \geq \left\lceil \frac{n+1}{2} \right\rceil. \quad (1.1)$$

Figure 1 shows the smallest interesting case.

The question is Problem 1183 in Bloom's collection of Erdős problems [Bl26], where it is listed as open. As recorded there, Erdős wrote that they had no plausible conjecture for the true order of magnitude, but asked whether

$$F(n) \geq n^{\omega(n)} \text{ for some } \omega(n) \rightarrow \infty, \quad \text{and} \quad F(n) < (1 + o(1))^n. \quad (1.2)$$

He also noted that Howorka had proved the first statement for colourings in which the colour of a set depends only on its size. We prove both statements in (1.2) for arbitrary colourings.

1.1. *Results.* The main result is a polynomial lower bound of every degree.

THEOREM 1.1. *For every integer $t \geq 1$ there is a constant $c_t > 0$ such that every 2-colouring of $2^{[n]}$ contains a monochromatic union-closed family with at least $c_t n^{t+1}$ members; that is,*

$$F(n) \geq c_t n^{t+1} \quad (n \geq 1).$$

One may take $c_t = (2t^t(t+2)^M(4M^2)^{t+1})^{-1}$, where $M = M(t)$ is the number defined in (3.1).

The upper bound comes from a random colouring and the Sauer–Shelah lemma.

THEOREM 1.2. *Let $n \geq 1$, and let $d \geq 1$ be an integer with $2^d > nd + 2$. Then*

$$F(n) \leq \sum_{j=0}^{d-1} \binom{n}{j}.$$

In particular, for $n \geq 32$,

$$F(n) \leq (\log_2 n + \log_2 \log_2 n + 2) n^{\log_2 n + \log_2 \log_2 n + 1} = n^{(1+o(1)) \log_2 n}.$$

Together they settle both parts of (1.2), and they show that the exponent $\omega(n)$ can never exceed $(1 + o(1)) \log_2 n$.

- COROLLARY 1.3. (i) *There is a function $\omega : \mathbb{N} \rightarrow \mathbb{R}$ with $\omega(n) \rightarrow \infty$ such that $F(n) \geq n^{\omega(n)}$ for every $n \geq 1$.*
(ii) *There is a function $\varepsilon : \mathbb{N} \rightarrow \mathbb{R}$ with $\varepsilon(n) \rightarrow 0$ such that $F(n) < (1 + \varepsilon(n))^n$ for every $n \geq 1$.*
(iii) *Every function ω as in (i) satisfies $\omega(n) \leq (1 + o(1)) \log_2 n$.*

The two functions of Erdős and Ulam turn out to be of quite different sizes. For sublattices the trivial bound (1.1) is correct up to a factor of order $(\log n)^2$.

THEOREM 1.4. *For every $n \geq 1$,*

$$\left\lceil \frac{n+1}{2} \right\rceil \leq f(n) \leq n^2 + n + 1 \quad \text{and} \quad f(n) < 12n (\log_2(n+2))^2.$$

Thus $f(n) = n^{1+o(1)}$, whereas $F(n)/n^C \rightarrow \infty$ for every constant C .

For small n we have computed both functions exactly; $f(n) = \lceil (n+1)/2 \rceil$ for $n \leq 10$, and the values $F(1), \dots, F(7)$ are 1, 2, 2, 3, 4, 5, 7 (Section 8).

1.2. *Colourings by cardinality and Hilbert cubes.* Howorka's case of the problem is arithmetic in nature. Call χ a *cardinality colouring* if $\chi(S) = g(|S|)$ for some $g : \{0, 1, \dots, n\} \rightarrow \{0, 1\}$. The building blocks of our families are the *t-cubes* $\{U \cup X_I : I \subseteq [t]\}$, where $U, X_1, \dots, X_t$ are pairwise disjoint, the X_i are nonempty and $X_I = \bigcup_{i \in I} X_i$. The sizes of the members of such a cube form the *Hilbert cube*

$$H(u; a_1, \dots, a_t) = \left\{ u + \sum_{i \in I} a_i : I \subseteq [t] \right\}, \quad u = |U|, \quad a_i = |X_i|,$$

so a cube is monochromatic under a cardinality colouring exactly when g is constant on the corresponding Hilbert cube. Hilbert's lemma [Hi1892], usually

regarded as the first result of Ramsey theory, says that every finite colouring of the positive integers has monochromatic Hilbert cubes of every dimension. Read through this dictionary, the proof of Theorem 1.1 produces many monochromatic Hilbert cubes spread along a single sequence of bounded steps.

THEOREM 1.5. *Let $t \geq 1$ and $M = M(t)$, let $n \geq 4M(t+1)$, and put $m = \lfloor (n-1)/M \rfloor$ and $s = \lfloor m/2 \rfloor$. For every $g : \{0, 1, \dots, n\} \rightarrow \{0, 1\}$ there are integers $\beta_1, \dots, \beta_m \in \{1, \dots, M\}$ and a colour c with the following property. For at least*

$$\frac{n^{t+1}}{2^t (4M^2)^{t+1}}$$

of the pairs (j, A) with $s < j \leq m$, $A \subseteq \{1, \dots, s\}$ and $|A| = t$, the function g takes the value c at every point of the Hilbert cube

$$H_\beta(j, A) = \left\{ \beta_1 + \beta_2 + \dots + \beta_j - \sum_{b \in B} \beta_b : B \subseteq A \right\}.$$

Different pairs (j, A) may give the same set of integers; there are at most $(n+1)M^t$ Hilbert cubes $H(u; a_1, \dots, a_t)$ in $\{0, \dots, n\}$ with all $a_i \leq M$. What Theorem 1.5 provides is one sequence β for which a fixed proportion of the cubes indexed by the pairs (j, A) have a common colour, all of them hanging from the partial sums $\beta_1 + \dots + \beta_j$. This is the structure that lets the corresponding families of sets be glued into one union-closed family. Section 6 also gives a short proof of Howorka's theorem from van der Waerden's theorem.

1.3. The idea of the proof of Theorem 1.1. Chains and cubes are monochromatic union-closed families that are easy to find, but neither is large. A chain has at most $n+1$ members, and a count of cubes shows that a random colouring has no monochromatic cube with more than $O(n \log \log n)$ members. The families we build are unions of many cubes that fit together. Take pairwise disjoint nonempty blocks $Y_1, \dots, Y_m$, put $C_j = Y_1 \cup \dots \cup Y_j$, and for $j > s$ and $A \subseteq [s]$ consider the cube $Q(j, A) = \{C_j \setminus Y_B : B \subseteq A\}$. The union of a member of $Q(j, A)$ and a member of $Q(j', A')$ with $j \leq j'$ is again a member of $Q(j', A')$, so the union of any collection of cubes $Q(j, A)$ that are monochromatic in a common colour is union-closed. There are about m^{t+1} pairs (j, A) with $|A| = t$, and it is therefore enough to make a fixed proportion of these cubes monochromatic.

For a fixed chain of blocks this cannot be arranged, since the colouring may have been chosen against it. We therefore randomise twice: the ground set is permuted by a uniformly random σ , and each block size is chosen at random from $\{1, \dots, M\}$. After a random permutation, the probability that a cube is monochromatic depends only on the sizes of its parts (Lemma 2.2). A

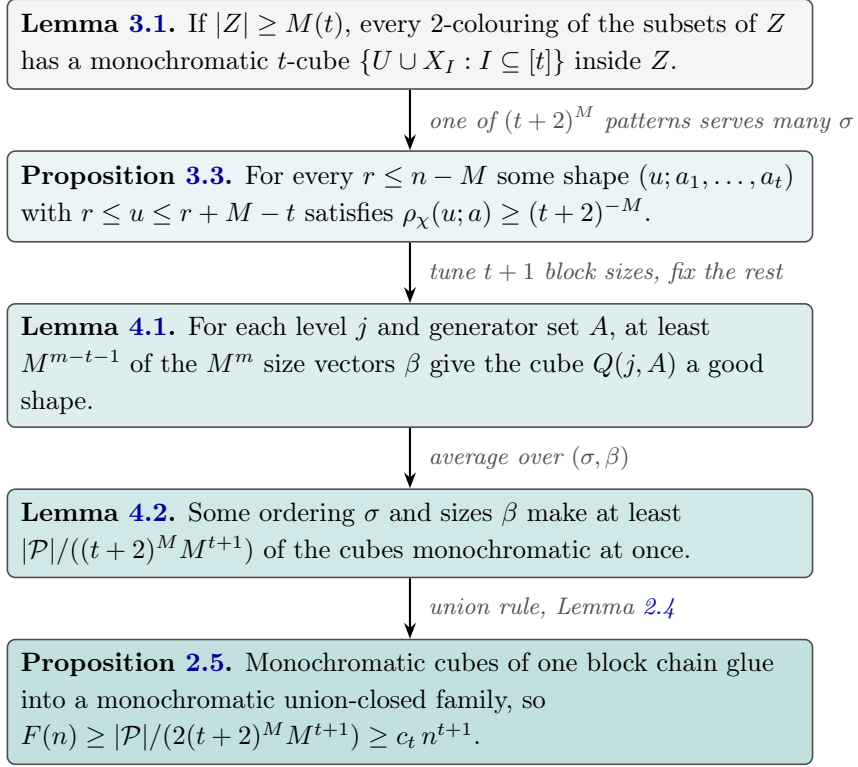


Figure 2. The structure of the proof of Theorem 1.1. Here $\mathcal{P}$ is the set of pairs (j, A) with $s < j \leq m$, $A \subseteq [s]$ and $|A| = t$, and ρ_χ is defined after Lemma 2.2.

pigeonhole argument over permutations, combined with the two-letter case of the multidimensional Hales–Jewett theorem, shows that for every r some shape whose base size lies between r and $r + M$ has probability at least $(t+2)^{-M}$ of being monochromatic (Proposition 3.3). The shape of $Q(j, A)$ is governed by $t+1$ block sizes, those of Y_j and of Y_a for $a \in A$, and these can be tuned independently of all the others (Lemma 4.1). Averaging over σ and the block sizes then produces one chain of blocks in which a proportion $(t+2)^{-M} M^{-t-1}$ of all the cubes is monochromatic (Lemma 4.2). Figure 2 shows how the pieces are assembled.

1.4. *Related work.* The union-closed families in this paper should not be confused with those of Frankl’s union-closed sets conjecture, which asks whether every union-closed family other than $\{\emptyset\}$ has an element lying in at least half of its members. That problem concerns the internal structure of one family and involves no colouring. Gilmer [Gi22] proved that some element lies in a

constant proportion of the members, and Alweiss, Huang and Sellke [AHS24] improved the constant. The entropy method behind these results compares the union of two independent random members of a family with a single random member, and we see no analogue of it here, where the family is not given in advance.

Families of 2^d sets of the form $\{U \cup X_I\}$ are called d -dimensional Boolean algebras by Gunderson, Rödl and Sidorenko [GRS99], who studied their extremal theory alongside that of d -dimensional affine cubes of integers and complete d -partite d -uniform hypergraphs. The passage between cubes of sets and Hilbert cubes of integers used in Section 6 is of the same kind. The cube lemma of Section 3 is the two-letter case of the multidimensional form of the Hales–Jewett theorem [HJ63]; see [GRS90] for this form and for the background in Ramsey theory, including van der Waerden’s theorem [vdW27]. The upper bound rests on the Sauer–Shelah lemma [Sa72, Sh72], which is also at the root of the theory of Vapnik–Chervonenkis dimension [VC71].

1.5. *Formal verification.* The definitions of f and F , the lower bound (1.1), Theorem 1.2 in the form $F(n) \leq \sum_{j < d} \binom{n}{j}$, Theorem 1.1 with a constant of the same kind, and both statements of (1.2) have been checked in the Lean proof assistant [MU21] using the Mathlib library [Mat20]. In the formal version the two conjectures are stated directly in the form (1.2), as in Corollary 1.3(i) and (ii). Section 9 lists what is covered and what is not.

1.6. *Organisation.* Section 2 introduces cubes and block chains and proves that monochromatic cubes of one chain can be glued. Section 3 proves the cube lemma and its supersaturated form, and Section 4 completes the proof of Theorem 1.1. Section 5 proves Theorem 1.2 and Corollary 1.3. Section 6 treats cardinality colourings, Section 7 the lattice function f , and Section 8 the exact values for small n . Section 9 describes the formalisation, and Section 10 collects open questions.

2. Cubes and block chains

Throughout, χ is a 2-colouring of $2^{[n]}$ and S_n is the group of permutations of $[n]$. A permutation acts on sets and on families of sets in the obvious way: $\sigma(S) = \{\sigma(x) : x \in S\}$ and $\sigma(\mathcal{F}) = \{\sigma(S) : S \in \mathcal{F}\}$. For sets $X_1, X_2, \dots$ and a finite set I of indices we write $X_I = \bigcup_{i \in I} X_i$, so that $X_\emptyset = \emptyset$.

Definition 2.1. Let $t \geq 0$. A t -cube in $2^{[n]}$ is a family

$$Q(U; X_1, \dots, X_t) = \{U \cup X_I : I \subseteq [t]\},$$

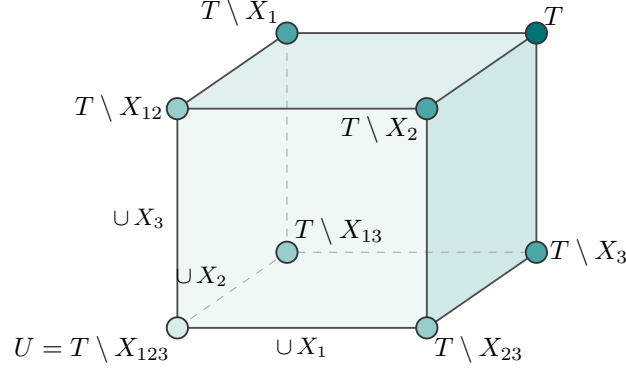


Figure 3. A 3-cube with top T and generators X_1, X_2, X_3 , where $X_{12} = X_1 \cup X_2$ and so on. Moving along an edge in direction i adds or removes X_i . The vertex $U = T \setminus X_{123}$ is the base.

where $U, X_1, \dots, X_t \subseteq [n]$ are pairwise disjoint and $X_1, \dots, X_t$ are nonempty. We call U the *base*, $T = U \cup X_{[t]}$ the *top* and $X_1, \dots, X_t$ the *generators*. The *shape* of the cube is the tuple $(|U|; |X_1|, \dots, |X_t|)$.

The 2^t sets $U \cup X_I$ are distinct, and

$$(U \cup X_I) \cup (U \cup X_J) = U \cup X_{I \cup J}, \quad (U \cup X_I) \cap (U \cup X_J) = U \cup X_{I \cap J},$$

so a t -cube is a sublattice of $2^{[n]}$ isomorphic to $2^{[t]}$. Writing $U \cup X_I = T \setminus X_{[t] \setminus I}$ describes the same cube from the top downwards, as the family $\{T \setminus X_I : I \subseteq [t]\}$; this is the form in which cubes appear in block chains. Figure 3 shows a 3-cube in this form.

LEMMA 2.2. *If Q and Q' are t -cubes in $2^{[n]}$ of the same shape, then*

$$|\{\sigma \in S_n : \sigma(Q) \text{ is monochromatic}\}| = |\{\sigma \in S_n : \sigma(Q') \text{ is monochromatic}\}|.$$

Proof. Let $Q = Q(U; X_1, \dots, X_t)$ and $Q' = Q(U'; X'_1, \dots, X'_t)$ with tops T and T' . The sets $U, X_1, \dots, X_t, [n] \setminus T$ partition $[n]$, as do $U', X'_1, \dots, X'_t, [n] \setminus T'$, and corresponding parts have the same size. So there is $\tau \in S_n$ with $\tau(U) = U'$ and $\tau(X_i) = X'_i$ for every i , and then $\tau(Q) = Q'$. Now $\sigma(Q') = (\sigma\tau)(Q)$, and $\sigma \mapsto \sigma\tau$ is a bijection of S_n . $\square$

For integers $u \geq 0$ and $a_1, \dots, a_t \geq 1$ with $u + a_1 + \dots + a_t \leq n$ we may therefore define

$$\rho_\chi(u; a_1, \dots, a_t) = \frac{1}{n!} |\{\sigma \in S_n : \sigma(Q) \text{ is monochromatic}\}|,$$

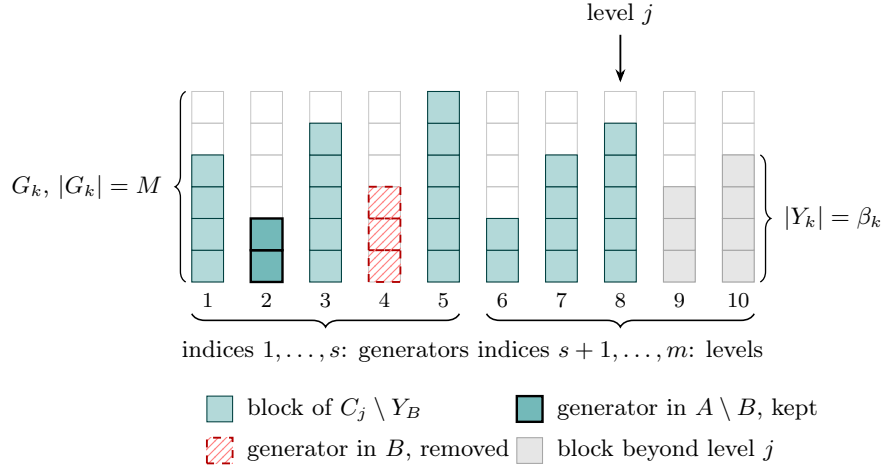


Figure 4. A block chain with $m = 10$ blocks placed in slots $G_1, \dots, G_{10}$ of size $M = 6$, with $|Y_k| = \beta_k$ and $\beta = (4, 2, 5, 3, 6, 2, 4, 5, 3, 4)$. Here $s = 5$, and the teal blocks form the member $C_8 \setminus Y_{\{4\}}$ of the cube $Q(8, \{2, 4\})$; the hatched block Y_4 is removed and the grey blocks lie beyond C_8 .

where Q is any t -cube of shape $(u; a_1, \dots, a_t)$. Since $\sigma(Q)$ is a uniformly random cube of that shape when σ is uniform, $\rho_\chi(u; a)$ is the probability that a random cube of shape $(u; a)$ is monochromatic.

Definition 2.3. Let $Y_1, \dots, Y_m \subseteq [n]$ be pairwise disjoint nonempty sets, the *blocks*, and put $C_j = Y_1 \cup \dots \cup Y_j$ for $0 \leq j \leq m$. For $1 \leq j \leq m$ and $A \subseteq [j-1]$ let

$$Q(j, A) = \{C_j \setminus Y_B : B \subseteq A\}.$$

This is the $|A|$ -cube with top C_j , generators Y_a ($a \in A$) and base $C_j \setminus Y_A$; its shape is $(\sum_{k \leq j, k \notin A} |Y_k|; (|Y_a|)_{a \in A})$, the generators being listed in increasing order of a .

Figure 4 shows a member of such a cube. The chain structure enters through the following rule, illustrated in Figure 5.

LEMMA 2.4 (Union rule). *Let $1 \leq j \leq j' \leq m$ and $B, B' \subseteq [j]$. Then*

$$(C_j \setminus Y_B) \cup (C_{j'} \setminus Y_{B'}) = C_{j'} \setminus Y_{B \cap B'}.$$

Proof. Both sides are contained in $C_{j'}$. Let $x \in C_{j'}$. If $x \notin Y_{B'}$, then x lies in $C_{j'} \setminus Y_{B'}$ and also outside $Y_{B \cap B'}$. Otherwise $x \in Y_b$ for some $b \in B'$; then $b \leq j$, so $x \in C_j$, and x lies in the left side if and only if $b \notin B$, that is, if and only if $x \notin Y_{B \cap B'}$. $\square$

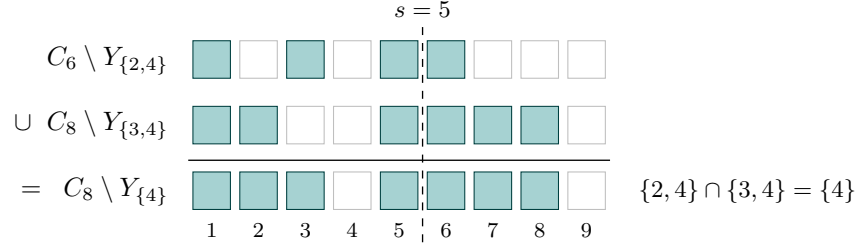


Figure 5. The union rule with $s = 5$: $C_6 \setminus Y_{\{2,4\}}$ and $C_8 \setminus Y_{\{3,4\}}$ have union $C_8 \setminus Y_{\{4\}}$. Filled squares are blocks present in the set.

PROPOSITION 2.5 (Gluing). *Let $1 \leq s < m$, and let $\mathcal{S}$ be a set of pairs (j, A) with $s < j \leq m$ and $A \subseteq [s]$ such that the cube $Q(j, A)$ is monochromatic for every $(j, A) \in \mathcal{S}$. Then there is a monochromatic union-closed family with at least $|\mathcal{S}|/2$ members.*

Proof. Choose a colour c such that the set $\mathcal{S}_c$ of pairs in $\mathcal{S}$ whose cube has colour c satisfies $|\mathcal{S}_c| \geq |\mathcal{S}|/2$, and let $\mathcal{F}$ be the union of the cubes $Q(j, A)$ over $(j, A) \in \mathcal{S}_c$. Every member of $\mathcal{F}$ has colour c .

Let $C_j \setminus Y_B \in Q(j, A)$ and $C_{j'} \setminus Y_{B'} \in Q(j', A')$ be members of $\mathcal{F}$, where $(j, A), (j', A') \in \mathcal{S}_c$ and, say, $j \leq j'$. As $B, B' \subseteq [s] \subseteq [j]$, Lemma 2.4 shows that their union is $C_{j'} \setminus Y_{B \cap B'}$, and since $B \cap B' \subseteq A'$ this set belongs to $Q(j', A') \subseteq \mathcal{F}$. So $\mathcal{F}$ is union-closed.

Finally, the map $(j, A) \mapsto C_j \setminus Y_A$ from $\mathcal{S}_c$ to $\mathcal{F}$ is injective. Indeed, $j > s$ means $j \notin A$, so $C_j \setminus Y_A$ meets Y_j and no block of larger index, which determines j ; and then A is the set of indices $k \leq s$ for which Y_k is disjoint from $C_j \setminus Y_A$. Hence $|\mathcal{F}| \geq |\mathcal{S}_c| \geq |\mathcal{S}|/2$. $\square$

The family $\mathcal{F}$ is not closed under intersection in general: the intersection of the two sets above is $C_j \setminus Y_{B \cup B'}$, and $B \cup B'$ need not be contained in A . This is as it must be, in view of Theorem 1.4.

3. The cube lemma and supersaturation

Define integers $M(t)$ by

$$M(0) = 0, \quad M(t) = M(t-1) + 2^{2^{M(t-1)}} \quad (t \geq 1), \quad (3.1)$$

so that $M(1) = 2$, $M(2) = 18$ and $M(3) = 18 + 2^{2^{18}}$. Note that $M(t) \geq t + 1$ for $t \geq 1$.

LEMMA 3.1 (Cube lemma). *Let $t \geq 0$ and let Z be a finite set with $|Z| \geq M(t)$. Every 2-colouring of 2^Z is constant on some t -cube $Q(U; X_1, \dots, X_t)$ with $U, X_1, \dots, X_t \subseteq Z$.*

Proof. For $t = 0$ a single set, say $U = \emptyset$, will do. Let $t \geq 1$, put $q = M(t-1)$ and $N = 2^{2^q}$, and choose disjoint sets $P, R \subseteq Z$ with $|P| = N$ and $|R| = q$, which is possible as $|Z| \geq q + N$. List $P = \{p_1, \dots, p_N\}$ and put $P_i = \{p_1, \dots, p_i\}$ for $0 \leq i \leq N$. Each i gives a colouring φ_i of 2^R , namely $\varphi_i(S) = \chi(P_i \cup S)$. There are only N colourings of 2^R and $N+1$ indices, so $\varphi_a = \varphi_b$ for some $a < b$. By induction φ_a is constant, with value c say, on a $(t-1)$ -cube $Q(U'; X_1, \dots, X_{t-1})$ inside R . Put $U = P_a \cup U'$ and $X_t = P_b \setminus P_a \neq \emptyset$. The sets $U, X_1, \dots, X_t$ are pairwise disjoint, and for $I \subseteq [t-1]$

$$\begin{aligned}\chi(U \cup X_I) &= \varphi_a(U' \cup X_I) = c, \\ \chi(U \cup X_I \cup X_t) &= \varphi_b(U' \cup X_I) = \varphi_a(U' \cup X_I) = c.\end{aligned}$$

So χ is constant on $Q(U; X_1, \dots, X_t)$. $\square$

Remark 3.2. Identify a set $S \subseteq Z$ with the word in $\{0, 1\}^Z$ that has the letter 1 exactly at the positions in S . A t -cube then becomes a t -dimensional combinatorial subspace: the positions in U carry the fixed letter 1, those outside T the fixed letter 0, and those in X_i the i -th variable. Lemma 3.1 is therefore the multidimensional form of the Hales–Jewett theorem [HJ63] for the alphabet $\{0, 1\}$, with the bound (3.1). The formal proof described in Section 9 uses the version of the Hales–Jewett theorem in Mathlib, which covers every finite alphabet and gives some M without an explicit value; nothing else in the argument depends on the value of M .

A single monochromatic cube is of little use by itself. What the proof needs is a shape that is monochromatic with probability bounded below independently of n , and this is obtained by applying Lemma 3.1 once for every permutation.

PROPOSITION 3.3 (Supersaturation). *Let $t \geq 1$ and $M = M(t)$, and let $r \geq 0$ be an integer with $r + M \leq n$. There are integers u and $a_1, \dots, a_t \geq 1$ with*

$$r \leq u, \quad (u - r) + a_1 + \dots + a_t \leq M, \quad \rho_\chi(u; a_1, \dots, a_t) \geq (t+2)^{-M}.$$

In particular $u \leq r + M - t$ and $a_i \leq M$ for every i .

Proof. Let $Z = [M]$ and $R = \{M+1, \dots, M+r\}$. For $\sigma \in S_n$ define a colouring of 2^Z by $S \mapsto \chi(\sigma(S \cup R))$. Lemma 3.1 gives disjoint $U, X_1, \dots, X_t \subseteq Z$ with the X_i nonempty such that this colouring is constant

on $Q(U; X_1, \dots, X_t)$; call $(U; X_1, \dots, X_t)$ a *pattern* for σ . A pattern is determined by recording, for each point of Z , whether it lies in U , in one of the t generators, or in none of them, so there are at most $(t+2)^M$ patterns. Some pattern $(U; X_1, \dots, X_t)$ is therefore a pattern for at least $n!/(t+2)^M$ permutations. For each such σ the family $\sigma(Q(U \cup R; X_1, \dots, X_t))$ is monochromatic, so the shape $u = |U| + r$, $a_i = |X_i|$ has $\rho_\chi(u; a) \geq (t+2)^{-M}$. The constraints on u and the a_i hold because $U, X_1, \dots, X_t$ are disjoint subsets of Z and the X_i are nonempty. $\square$

The lower bound $(t+2)^{-M}$ depends on t alone. The price paid for it is that the base size u is only known to lie in a window of length M above r ; the block chains of the next section are designed to absorb exactly this uncertainty.

4. Proof of Theorem 1.1

Fix $t \geq 1$, put $M = M(t)$, and let n be large enough that

$$m = \left\lfloor \frac{n-1}{M} \right\rfloor \quad \text{and} \quad s = \left\lfloor \frac{m}{2} \right\rfloor$$

satisfy $s \geq t$. Note that $mM \leq n-1$. Fix pairwise disjoint sets $G_1, \dots, G_m \subseteq [n]$ of size M , the *slots*, and list each slot as $G_k = \{g_{k,1}, \dots, g_{k,M}\}$; for instance $g_{k,i} = (k-1)M + i$. For a vector $\beta \in [M]^m$ of block sizes let

$$Y_k^\beta = \{g_{k,1}, \dots, g_{k,\beta_k}\} \quad (1 \leq k \leq m),$$

and for $\sigma \in S_n$ let $Q_{\sigma,\beta}(j, A)$ denote the cube $Q(j, A)$ of Definition 2.3 built from the blocks $\sigma(Y_1^\beta), \dots, \sigma(Y_m^\beta)$. Thus $Q_{\sigma,\beta}(j, A) = \sigma(Q_{\text{id},\beta}(j, A))$, and $Q_{\sigma,\beta}(j, A)$ has shape

$$\left(\sum_{k \leq j, k \notin A} \beta_k; (\beta_a)_{a \in A} \right). \quad (4.1)$$

Let $\mathcal{P}$ be the set of pairs (j, A) with $s < j \leq m$, $A \subseteq [s]$ and $|A| = t$. Call β *good for* (j, A) if the shape (4.1) satisfies $\rho_\chi \geq (t+2)^{-M}$.

LEMMA 4.1 (Slices). *Let $1 \leq j \leq m$ and $A \subseteq [j-1]$ with $|A| = t$. At least M^{m-t-1} vectors $\beta \in [M]^m$ are good for (j, A) .*

Proof. Write $A = \{a_{(1)} < \dots < a_{(t)}\}$ and let $K = [m] \setminus (A \cup \{j\})$, a set of $m-t-1$ indices. We show that every choice of $(\beta_k)_{k \in K} \in [M]^K$ extends to a vector β that is good for (j, A) . Different choices give different vectors, which proves the lemma.

Fix $(\beta_k)_{k \in K}$, let $R = \sum_{k < j, k \notin A} \beta_k$, which involves only indices in K , and put $r = R + 1$. Since $R \leq (j-1-t)M \leq (m-2)M$, we have $r + M \leq (m -$

$1)M + 1 \leq n$. Proposition 3.3 gives u and $a_1, \dots, a_t$ with $\rho_\chi(u; a) \geq (t+2)^{-M}$, where $z = u - r$ satisfies $0 \leq z \leq M - t \leq M - 1$ and $1 \leq a_i \leq M$. Set

$$\beta_j = z + 1, \quad \beta_{a_{(i)}} = a_i \quad (1 \leq i \leq t).$$

These values lie in $[M]$, and the shape (4.1) becomes $(R + z + 1; a_1, \dots, a_t) = (u; a_1, \dots, a_t)$. So β is good for (j, A) . $\square$

LEMMA 4.2 (Averaging). *There are $\sigma \in S_n$ and $\beta \in [M]^m$ such that at least*

$$\frac{|\mathcal{P}|}{(t+2)^M M^{t+1}}$$

of the cubes $Q_{\sigma, \beta}(j, A)$, $(j, A) \in \mathcal{P}$, are monochromatic.

Proof. Count the triples $(\sigma, \beta, (j, A)) \in S_n \times [M]^m \times \mathcal{P}$ for which $Q_{\sigma, \beta}(j, A)$ is monochromatic. For fixed $(j, A) \in \mathcal{P}$ and β good for (j, A) , the number of suitable σ is $n!$ times the value of ρ_χ at the shape (4.1), which is at least $n! (t+2)^{-M}$. By Lemma 4.1 the number of triples is at least $|\mathcal{P}| M^{m-t-1} n! (t+2)^{-M}$. There are $n! M^m$ pairs (σ, β) , so one of them occurs in at least $|\mathcal{P}| (t+2)^{-M} M^{-t-1}$ triples. $\square$

Proof of Theorem 1.1. Let c_t be as in the statement. Since $M \geq t+1$ we have $c_t \leq (4M(t+1))^{-(t+1)}$, so for $n < 4M(t+1)$ the bound $F(n) \geq 1 > c_t n^{t+1}$ is trivial. Let $n \geq 4M(t+1)$.

Take σ, β as in Lemma 4.2 and apply Proposition 2.5 to the blocks $\sigma(Y_1^\beta), \dots, \sigma(Y_m^\beta)$, which are pairwise disjoint and nonempty, and to the set $\mathcal{S} \subseteq \mathcal{P}$ of pairs whose cube is monochromatic. Every colouring therefore has a monochromatic union-closed family with at least

$$\frac{|\mathcal{P}|}{2(t+2)^M M^{t+1}} \tag{4.2}$$

members. It remains to bound $|\mathcal{P}| = (m-s) \binom{s}{t}$ from below. We have $m \geq (n-M)/M$, so $s \geq (m-1)/2 \geq n/(2M) - 1 \geq n/(4M)$ because $n \geq 4M$. In particular $s \geq t+1$, so $\binom{s}{t} \geq (s/t)^t$, and $m-s \geq s$. Hence

$$|\mathcal{P}| \geq \frac{s^{t+1}}{t^t} \geq \frac{n^{t+1}}{t^t (4M)^{t+1}},$$

and (4.2) is at least $c_t n^{t+1}$. $\square$

Remark 4.3. The proof gives an explicit, if very slowly growing, function in Corollary 1.3(i). By Theorem 1.1, $F(n) \geq n^t$ as soon as $n \geq 1/c_t$. For $t = 1$ we have $1/c_1 = 4608 < 2^{16}$, and for $t \geq 2$ a crude estimate using $t+1 \leq M(t)$ and $M(t) \geq 18$ gives $\log_2(1/c_t) \leq 1 + 2M + 4M \log_2(M+1) \leq 2^{M+2}$ with $M = M(t)$. So $F(n) \geq n^t$ whenever $\log_2 \log_2 n \geq M(t) + 2$. Let $E(x) = 2^x$,

let E^k be its k -fold iterate, and let $\log^* n$ be the least k with $E^k(0) \geq n$. Since $x + 2^{2^x} \leq E^3(x)$ for $x \geq 0$, (3.1) gives $M(t) \leq E^{3t}(0)$, and the condition above holds once $n \geq E^{3t+3}(0)$. Consequently the function ω constructed in the proof of Corollary 1.3(i) satisfies $\omega(n) \geq \frac{1}{3} \log^* n - 2$. We have made no attempt to improve this, and it is very far from the limit $(1 + o(1)) \log_2 n$ of Corollary 1.3(iii).

5. The upper bound

A family $A_1, \dots, A_d$ of subsets of $[n]$ is an *independent system* if each A_i has a *private element*, that is, some $x_i \in A_i$ lying in no $A_{i'}$ with $i' \neq i$. A set $X \subseteq [n]$ is *shattered* by a family $\mathcal{F}$ if $\{A \cap X : A \in \mathcal{F}\} = 2^X$, and the *Vapnik–Chervonenkis dimension* $\text{vc}(\mathcal{F})$ is the largest size of a shattered set.

- LEMMA 5.1. (a) *If $A_1, \dots, A_d$ is an independent system, the $2^d - 1$ unions $\bigcup_{i \in I} A_i$, $\emptyset \neq I \subseteq [d]$, are pairwise distinct.*
 (b) *If $\mathcal{F}$ shatters a set of size d , then $\mathcal{F}$ contains an independent system of d sets.*
 (c) (Sauer [Sa72], Shelah [Sh72]) *If $\text{vc}(\mathcal{F}) \leq d$, then $|\mathcal{F}| \leq \sum_{j=0}^d \binom{n}{j}$.*

Proof. (a) The private element x_i lies in $\bigcup_{i' \in I} A_{i'}$ exactly when $i \in I$, so the union determines I . (b) If $X = \{x_1, \dots, x_d\}$ is shattered, choose $A_i \in \mathcal{F}$ with $A_i \cap X = \{x_i\}$; then x_i is a private element of A_i . (c) This is the Sauer–Shelah lemma. $\square$

Proof of Theorem 1.2. Let χ be a uniformly random 2-colouring of $2^{[n]}$, the colours of the 2^n sets being independent and uniform. Call an ordered d -tuple $(A_1, \dots, A_d)$ of subsets of $[n]$ *bad* if it is an independent system and the $2^d - 1$ sets $\bigcup_{i \in I} A_i$, $\emptyset \neq I \subseteq [d]$, all have the same colour. By Lemma 5.1(a) these sets are distinct, so a fixed independent tuple is bad with probability $2^{1-(2^d-1)} = 2^{2-2^d}$. There are at most 2^{nd} tuples, and $2^{nd} \cdot 2^{2-2^d} < 1$ by hypothesis, so some colouring has no bad tuple. Fix it.

Let $\mathcal{F}$ be a monochromatic union-closed family. If $\mathcal{F}$ contained an independent system of d sets, all unions of its members would lie in $\mathcal{F}$ and have the same colour, and the system would be bad. So $\mathcal{F}$ contains no independent system of d sets, $\text{vc}(\mathcal{F}) \leq d - 1$ by Lemma 5.1(b), and $|\mathcal{F}| \leq \sum_{j < d} \binom{n}{j}$ by Lemma 5.1(c).

For the second assertion let $n \geq 32$ and $d = \lceil \log_2 n + \log_2 \log_2 n \rceil + 1$. Then $2^d \geq 2n \log_2 n$, while $nd + 2 \leq n(\log_2 n + \log_2 \log_2 n + 2) + 2$. The required inequality $2n \log_2 n > n \log_2 n + n \log_2 \log_2 n + 2n + 2$ is equivalent to $\log_2 n - \log_2 \log_2 n > 2 + 2/n$, which holds for $n \geq 32$: at $n = 32$ the

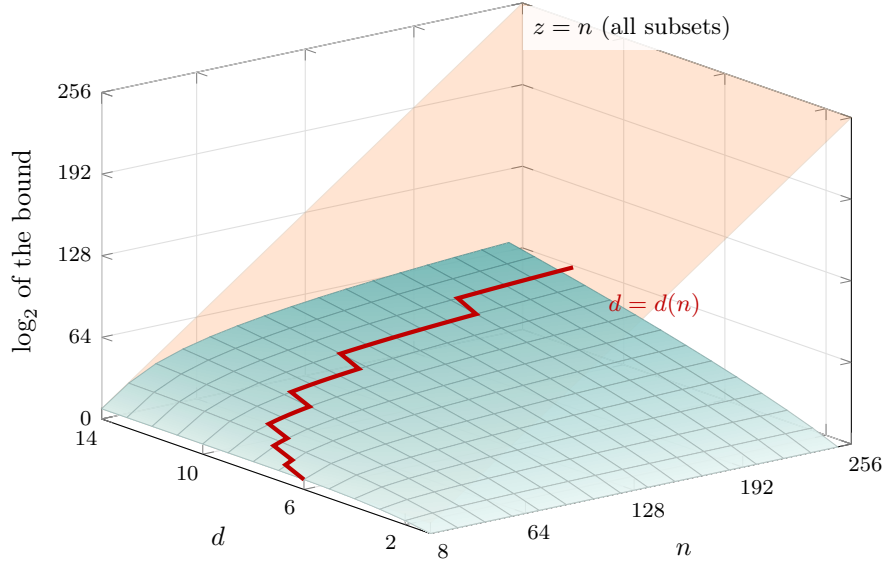


Figure 6. The surface $\log_2 \sum_{j < d} \binom{n}{j}$ over $8 \leq n \leq 256$ and $1 \leq d \leq 14$, with the plane $z = n$ of the trivial bound 2^n . The red curve follows the least admissible $d = d(n)$ of Theorem 1.2, along which the bound stays far below the plane.

left side is $5 - \log_2 5 > 2.67$, and it increases with n . Finally $\binom{n}{j} \leq n^j$, so $\sum_{j < d} \binom{n}{j} \leq d n^{d-1}$, and $d \leq \log_2 n + \log_2 \log_2 n + 2$. $\square$

Figure 6 shows the size of the bound of Theorem 1.2 as a function of n and d . Taking for d the least admissible value $d(n)$ gives a bound valid for every $n \geq 1$; for instance $d(n) = 6$ for $6 \leq n \leq 10$, and $d(n) = \log_2 n + \log_2 \log_2 n + O(1)$ as $n \rightarrow \infty$.

Proof of Corollary 1.3. (i) For $n \geq 1$ let $\omega(n)$ be the largest integer $c \in \{0, \dots, n\}$ with $n^c \leq F(n)$; it exists because $F(n) \geq 1$. Given $c \geq 1$, Theorem 1.1 with $t = c$ gives $F(n) \geq c n^{c+1} \geq n^c$ for all $n \geq \max(c, 1/c_c)$, and for these n we have $\omega(n) \geq c$. Hence $\omega(n) \rightarrow \infty$.

(ii) Put $\varepsilon(n) = F(n)^{1/n} - 1 + 1/n$. Then $(1 + \varepsilon(n))^n = (F(n)^{1/n} + 1/n)^n > F(n)$. By Theorem 1.2, $\log F(n) = O((\log n)^2)$, so $F(n)^{1/n} \rightarrow 1$ and $\varepsilon(n) \rightarrow 0$.

(iii) If $n^{\omega(n)} \leq F(n)$, Theorem 1.2 gives $\omega(n) \log_2 n \leq \log_2 F(n) \leq (1 + o(1))(\log_2 n)^2$. $\square$

The random colouring does contain large monochromatic union-closed families, and it is worth seeing how large, since this indicates how much room is left between the two theorems.

PROPOSITION 5.2. *Let χ be a uniformly random 2-colouring of $2^{[n]}$. The largest monochromatic union-closed family has expected size at least*

$$\sum_{j=0}^n \binom{n}{j} 2^{1-2^j} \geq \frac{2}{n} \left(\frac{n}{j_0}\right)^{j_0} \geq n^{\log_2 \log_2 n - 2 - o(1)}, \quad j_0 = \lfloor \log_2 \log_2 n \rfloor,$$

the inequality holding for $n \geq 4$.

Proof. Let $c = \chi([n])$ and let $\mathcal{Y}$ be the set of $Y \subseteq [n]$ such that $\chi([n] \setminus Y') = c$ for every $Y' \subseteq Y$. Then $\mathcal{Y}$ is closed under taking subsets and under intersection, so $\mathcal{F} = \{[n] \setminus Y : Y \in \mathcal{Y}\}$ is monochromatic and union-closed, because $([n] \setminus Y) \cup ([n] \setminus Y') = [n] \setminus (Y \cap Y')$. A set Y of size j lies in $\mathcal{Y}$ exactly when the $2^j - 1$ distinct sets $[n] \setminus Y'$ with $\emptyset \neq Y' \subseteq Y$ all have colour c , which happens with probability 2^{1-2^j} . Summing over Y gives $\mathbb{E}|\mathcal{F}|$. For $n \geq 4$ we have $j_0 \geq 1$ and $2^{j_0} \leq \log_2 n$, so the term $j = j_0$ is at least $(2/n) \binom{n}{j_0} \geq (2/n)(n/j_0)^{j_0}$, and $j_0^{j_0} = n^{o(1)}$. $\square$

So for the random colouring the exponent ω is at least about $\log_2 \log_2 n$, while Theorem 1.2, which is also proved with the random colouring, allows it to be as large as $\log_2 n$. We do not know where the truth lies (Question 10.1).

6. Cardinality colourings, Hilbert cubes and arithmetic progressions

In this section $\chi(S) = g(|S|)$ for a function $g : \{0, 1, \dots, n\} \rightarrow \{0, 1\}$. A t -cube of shape $(u; a_1, \dots, a_t)$ consists of sets whose sizes are the elements of $H(u; a_1, \dots, a_t)$, and every permutation preserves sizes, so

$$\rho_\chi(u; a_1, \dots, a_t) = \begin{cases} 1 & \text{if } g \text{ is constant on } H(u; a_1, \dots, a_t), \\ 0 & \text{otherwise.} \end{cases} \quad (6.1)$$

Figure 7 shows a monochromatic Hilbert cube of dimension two.

Applied to the colouring $S \mapsto g(r + |S|)$ of $2^{[M]}$, Lemma 3.1 gives the two-colour case of Hilbert's lemma in the following windowed form.

COROLLARY 6.1. *Let $t \geq 1$, $M = M(t)$ and $r \geq 0$. Every $g : \{r, r+1, \dots, r+M\} \rightarrow \{0, 1\}$ is constant on some Hilbert cube $H(u; a_1, \dots, a_t)$ with $a_i \geq 1$, $u \geq r$ and $u + a_1 + \dots + a_t \leq r + M$.*

Proof of Theorem 1.5. Consider the cardinality colouring $\chi(S) = g(|S|)$ and the construction of Section 4, whose hypotheses hold since $n \geq 4M(t+1)$ gives $s \geq n/(4M) \geq t+1$. By (6.1), a vector β is good for (j, A) precisely when g is constant on the Hilbert cube formed by the sizes of the members of

$H(u; a_1, a_2) = \{u, u + a_1, u + a_2, u + a_1 + a_2\}$ with $u = 5, a_1 = 3, a_2 = 7$

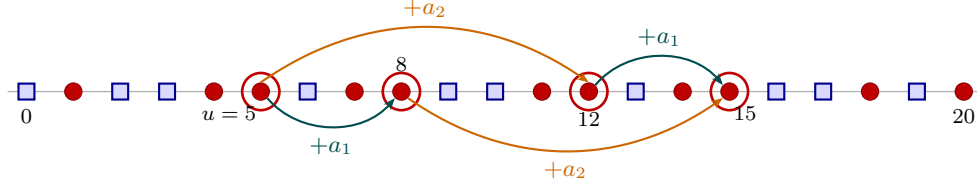


Figure 7. A colouring g of $\{0, 1, \dots, 20\}$, red discs and blue squares, and a red Hilbert cube $H(5; 3, 7) = \{5, 8, 12, 15\}$. For the cardinality colouring $\chi(S) = g(|S|)$, every 2-cube of shape $(5; 3, 7)$ is red.

$Q_{\text{id}, \beta}(j, A)$, and these sizes are

$$|C_j \setminus Y_B| = \beta_1 + \dots + \beta_j - \sum_{b \in B} \beta_b \quad (B \subseteq A),$$

that is, the elements of $H_\beta(j, A)$. By Lemma 4.1, for every $(j, A) \in \mathcal{P}$ at least M^{m-t-1} vectors β make $H_\beta(j, A)$ monochromatic. Averaging over the M^m vectors, as in Lemma 4.2 but without the permutations, some β makes at least $|\mathcal{P}|/M^{t+1}$ of the cubes $H_\beta(j, A)$ monochromatic, and at least half of these share a colour c . The estimate $|\mathcal{P}| \geq n^{t+1}/(t^t(4M)^{t+1})$ from the proof of Theorem 1.1 completes the proof. All elements of the cubes lie in $\{0, \dots, n\}$, since $\beta_1 + \dots + \beta_m \leq mM \leq n - 1$. $\square$

Given β and c as in Theorem 1.5, the cubes $Q_{\text{id}, \beta}(j, A)$ attached to the pairs of colour c are monochromatic under χ , and Proposition 2.5 glues them into a single union-closed family of colour c . For cardinality colourings the proof of Theorem 1.1 is therefore entirely a statement about sums of a bounded sequence of integers, and the factor $(t+2)^{-M}$ coming from permutations disappears.

There is also a direct route through arithmetic progressions, which gives Howorka's theorem with different constants. Let $W(k)$ be the van der Waerden number, the least N such that every 2-colouring of $\{1, \dots, N\}$ contains a monochromatic arithmetic progression of length k ; it is finite by van der Waerden's theorem [vdW27], and Shelah [Sh88] showed that it is bounded by a primitive recursive function of k .

PROPOSITION 6.2. *Suppose that g is constant on the arithmetic progression $a, a + e, \dots, a + (k-1)e$, where $e \geq 1, k \geq 1$ and $a + (k-1)e \leq n$. Put $q = \lfloor a/e \rfloor$ and $b = q + k - 1$. Then χ has a monochromatic union-closed family with $\sum_{i=0}^{k-1} \binom{b}{i}$ members and a monochromatic sublattice with 2^{k-1} members.*

Proof. Let $a_0 = a - qe$, so $0 \leq a_0 < e$, and choose pairwise disjoint sets $D, X_1, \dots, X_b \subseteq [n]$ with $|D| = a_0$ and $|X_i| = e$; this is possible because $a_0 + be = a + (k-1)e \leq n$. The family

$$\mathcal{F} = \{D \cup X_I : I \subseteq [b], |I| \geq q\}$$

is union-closed, since $|I \cup I'| \geq q$ whenever $|I| \geq q$. A member with index set I has size $a_0 + |I|e = a + (|I| - q)e$, where $0 \leq |I| - q \leq k-1$, so all members have the colour of the progression, and $|\mathcal{F}| = \sum_{i=q}^b \binom{b}{i} = \sum_{i=0}^{k-1} \binom{b}{i}$. For the sublattice take the $(k-1)$ -cube $Q(D'; X_1, \dots, X_{k-1})$ with $|D'| = a$ and $|X_i| = e$, whose member sizes $a + |I|e$ lie in the progression. $\square$

THEOREM 6.3 (Howorka). *Let $k \geq 2$ and $n \geq 2W(k)$. Every cardinality colouring of $2^{[n]}$ has a monochromatic union-closed family with at least*

$$\binom{\lfloor n/(2W(k)) \rfloor + k - 1}{k - 1} \geq \left(\frac{n}{2(k-1)W(k)} \right)^{k-1}$$

members. Consequently there is $\omega(n) \rightarrow \infty$ such that every cardinality colouring of $2^{[n]}$ has a monochromatic union-closed family with at least $n^{\omega(n)}$ members.

Proof. Let $W = W(k)$ and $m_0 = \lceil n/2 \rceil$. The integers $m_0, m_0+1, \dots, m_0+W-1$ lie in $\{0, \dots, n\}$ because $n \geq 2W$, so g is constant on a progression $a, a+e, \dots, a+(k-1)e$ among them. Here $a \geq n/2$ and $(k-1)e \leq W-1$, so $e < W$ and $\lfloor a/e \rfloor \geq \lfloor n/(2W) \rfloor$. Proposition 6.2 gives a monochromatic union-closed family with at least $\binom{b}{k-1}$ members, where $b \geq \lfloor n/(2W) \rfloor + k - 1 \geq n/(2W)$, and $\binom{b}{k-1} \geq (b/(k-1))^{k-1}$. For the last statement let $\omega(n) = k/3$, where k is the largest integer with $k \geq 4$ and $2(k-1)W(k) \leq n^{1/2}$, and let $\omega(n) = 0$ if there is no such k . The bound is then at least $n^{(k-1)/2} \geq n^{k/3}$. $\square$

The theorem is attributed to Howorka in [Bl26] without a reference, and as we have not found Howorka's proof in print we have included the short argument above. Proposition 6.2 also restricts the cardinality colourings that are extremal for f . If no monochromatic sublattice has more than $\lceil (n+1)/2 \rceil$ members, then g has no monochromatic progression of length k with $2^{k-1} > \lceil (n+1)/2 \rceil$. This condition was used in the search for the colouring g_{10} of Section 8.

7. The lattice function

The lower bound in Theorem 1.4 is (1.1). The upper bounds come from a random colouring and a count of sublattices, which in turn rests on the structure of a finite distributive lattice.

Let $\mathcal{L} \subseteq 2^{[n]}$ be a sublattice with $|\mathcal{L}| \geq 2$, and put $B_0 = \bigcap_{A \in \mathcal{L}} A$ and $T_0 = \bigcup_{A \in \mathcal{L}} A$, both members of $\mathcal{L}$. For $x \in T_0 \setminus B_0$ let $A_x = \bigcap \{A \in \mathcal{L} : x \in A\}$,

the least member of $\mathcal{L}$ containing x , and let $\mathcal{J} = \mathcal{J}(\mathcal{L}) = \{A_x : x \in T_0 \setminus B_0\}$, ordered by inclusion. A *down-set* of a poset P is a set $D \subseteq P$ with $y \leq x \in D \Rightarrow y \in D$; we write $\text{Down}(P)$ for the set of down-sets and $w(P)$ for the largest size of an antichain.

LEMMA 7.1. *Let $\mathcal{L}$ be a sublattice of $2^{[n]}$ with $|\mathcal{L}| \geq 2$.*

- (a) *Every $A \in \mathcal{L}$ equals $B_0 \cup \bigcup \{J \in \mathcal{J} : J \subseteq A\}$.*
- (b) *If $J \in \mathcal{J}$ and $J \subseteq A_1 \cup \dots \cup A_r$ with $A_1, \dots, A_r \in \mathcal{L}$, then $J \subseteq A_i$ for some i .*
- (c) *The map $A \mapsto \{J \in \mathcal{J} : J \subseteq A\}$ is a bijection from $\mathcal{L}$ onto $\text{Down}(\mathcal{J})$.*
- (d) *$1 \leq |\mathcal{J}| \leq n$ and $w(\mathcal{J}) \leq \log_2 |\mathcal{L}|$.*
- (e) *Let $\psi : [n] \rightarrow \mathcal{J} \cup \{\perp, \top\}$ be given by $\psi(x) = \perp$ for $x \in B_0$, $\psi(x) = \top$ for $x \notin T_0$, and $\psi(x) = A_x$ otherwise. Then $\mathcal{L}$ consists of the sets $\{x : \psi(x) = \perp \text{ or } \psi(x) \in D\}$, $D \in \text{Down}(\mathcal{J})$. So $\mathcal{L}$ is determined by the poset $\mathcal{J}$, given abstractly on a set of labels, together with ψ .*

Proof. (a) If $x \in A \setminus B_0$ then $A_x \subseteq A$, so A is contained in the right-hand side, and the reverse inclusion is clear.

(b) Let $J = A_x$. Then $x \in A_i$ for some i , and $A_x \subseteq A_i$ by minimality.

(c) The map is injective by (a), and its values are down-sets. Given a down-set D , the set $A_D = B_0 \cup \bigcup_{J \in D} J$ lies in $\mathcal{L}$. If $J \in \mathcal{J}$ and $J \subseteq A_D$, then by (b) either $J \subseteq B_0$, which is impossible because $J = A_x$ with $x \notin B_0$, or $J \subseteq J'$ for some $J' \in D$, and then $J \in D$. So A_D is mapped to D .

(d) As $|\mathcal{L}| \geq 2$ we have $T_0 \neq B_0$, so $\mathcal{J} \neq \emptyset$. List $\mathcal{J}$ as $J_1, \dots, J_k$ in an order compatible with inclusion. The sets $\{J_1, \dots, J_i\}$, $0 \leq i \leq k$, are down-sets, so by (c) they correspond to a strictly increasing chain of $k+1$ subsets of $[n]$, and $k \leq n$. If $S \subseteq \mathcal{J}$ is an antichain, the down-sets generated by the $2^{|S|}$ subsets of S are distinct, so $2^{|S|} \leq |\mathcal{L}|$.

(e) For $x \in T_0 \setminus B_0$ and $J \in \mathcal{J}$ we have $x \in J$ if and only if $A_x \subseteq J$. Hence $x \in A_D$ if and only if $A_x \subseteq J$ for some $J \in D$, which for a down-set D means $A_x \in D$. $\square$

Part (c) is Birkhoff's representation of finite distributive lattices [Bi37], specialised to sublattices of $2^{[n]}$.

LEMMA 7.2. *Let $k \geq 1$ and $w \geq 1$. The number of partial orders of width at most w on the labelled set $[k]$ is at most $(k+1)^{k(w+2)}$.*

Proof. Put $v = \min(\lfloor w \rfloor, k)$. The width of the poset is at most v , so by Dilworth's theorem [Di50] it is the union of v disjoint chains $\Gamma_1, \dots, \Gamma_v$, some possibly empty. Record for each $x \in [k]$ the index $c(x)$ of its chain,

its position $p(x)$ in that chain counted from the bottom, and for each $i \leq v$ the number $r_i(x)$ of elements of Γ_i below or equal to x . The elements of Γ_i below or equal to x form an initial segment of Γ_i , so $y \leq x$ holds exactly when $p(y) \leq r_{c(y)}(x)$. The data determine the order, and there are at most $v^k k^k (k+1)^{kv} \leq (k+1)^{k(v+2)} \leq (k+1)^{k(w+2)}$ possibilities. $\square$

LEMMA 7.3. *For $m \geq 2$ the number $N(m)$ of sublattices of $2^{[n]}$ with exactly m members satisfies $N(m) \leq (n+2)^{n(\log_2 m+4)}$. The number of sublattices with at least two members is at most $2^{n(n+1)}$.*

Proof. By Lemma 7.1(d),(e), a sublattice with m members is determined by a partial order of width at most $\log_2 m$ on a label set $[k]$ with $k \leq n$, together with a map $[n] \rightarrow [k] \cup \{\perp, \top\}$. Lemma 7.2 gives $N(m) \leq \sum_{k=1}^n (k+1)^{k(\log_2 m+2)} (k+2)^n \leq n(n+2)^{n(\log_2 m+3)} \leq (n+2)^{n(\log_2 m+4)}$. For the second statement, Lemma 7.1(a) shows that $\mathcal{L}$ consists of the unions of B_0 with subfamilies of $\mathcal{J}$, so $\mathcal{L}$ is determined by the family $\{B_0\} \cup \mathcal{J}$ of at most $n+1$ sets, in which B_0 is the smallest member. Such a family can be written as a sequence of exactly $n+1$ subsets of $[n]$, repeating a member if necessary, and there are $2^{n(n+1)}$ such sequences. $\square$

Proof of Theorem 1.4. Let χ be a uniformly random colouring. A fixed family of m sets is monochromatic with probability 2^{1-m} .

For the first bound put $m_1 = n^2 + n + 2$. By Lemma 7.3 the expected number of monochromatic sublattices with at least m_1 members is at most $2^{n(n+1)} 2^{1-m_1} = 1/2$, so some colouring has none, and $f(n) \leq m_1 - 1$.

For the second bound we may assume $n \geq 3$, since for $n \leq 2$ it is weaker than $f(n) \leq 2^n$. Put $\ell = \log_2(n+2)$, so $\ell > 2.32$, and $L = n\ell$ and $m_2 = \lceil 12n\ell^2 \rceil$. By Lemma 7.3, $N(m)2^{1-m} \leq 2^{L(\log_2 m+4)+1-m}$ for $m \geq 2$. The function $\varphi(m) = m/2 - L(\log_2 m + 4) - 1$ is increasing for $m > 2L/\ln 2$, and $m_2 > 2.9L > 2L/\ln 2$, so $\varphi(m) \geq \varphi(m_2)$ for $m \geq m_2$. Using $m_2 \leq 13n\ell^2$, $\log_2 n \leq \ell$ and $\log_2 \ell \leq \ell - 1$, which holds for $\ell \geq 2$, we get $\log_2 m_2 + 4 \leq \log_2 13 + 3\ell + 2 < 3\ell + 5.71$, hence

$$L(\log_2 m_2 + 4) + 1 \leq 3n\ell^2 + 5.71n\ell + 1 \leq 6n\ell^2 \leq m_2/2,$$

the middle inequality being $3\ell \geq 5.71 + 1/(n\ell)$, true for $\ell > 2.32$. So $\varphi(m_2) \geq 0$, and $N(m)2^{1-m} \leq 2^{-m/2}$ for all $m \geq m_2$. The expected number of monochromatic sublattices with at least m_2 members is then at most $\sum_{m \geq m_2} 2^{-m/2} = 2^{-m_2/2}/(1 - 2^{-1/2}) < 1$, and $f(n) \leq m_2 - 1 < 12n\ell^2$.

The last assertion of the theorem follows from these bounds and Theorem 1.1. $\square$

The first bound is the better one for n up to about 1300. Neither can be pushed below n by studying the random colouring alone.

PROPOSITION 7.4. *With probability greater than $1/2$, a uniformly random 2-colouring of $2^{[n]}$ has a monochromatic chain of n sets, and hence a monochromatic sublattice of n members.*

Proof. Build the chain greedily. Put $S_0 = \emptyset$ and $c = \chi(\emptyset)$. Given S_i with $|S_i| = i \leq n-2$ and colour c , the $n-i$ sets $S_i \cup \{x\}$, $x \notin S_i$, have not been looked at before, and their colours are independent and uniform. If one of them has colour c , call it S_{i+1} ; the step fails with probability $2^{-(n-i)}$. The probability that some step fails is at most $\sum_{i=0}^{n-2} 2^{-(n-i)} = \sum_{j=2}^n 2^{-j} < 1/2$, and otherwise $S_0 \subset S_1 \subset \dots \subset S_{n-1}$ is a monochromatic chain. $\square$

8. Exact values for small n

For small n both functions can be computed. Fix n and a target m . A colouring with no monochromatic closed family of at least m members (closed meaning closed under union for F , and under union and intersection for f) is a satisfying assignment of the Boolean formula in the variables x_S , $S \subseteq [n]$, that contains, for every closed family $\mathcal{F}$ with at least m members, the two clauses $\bigvee_{S \in \mathcal{F}} x_S$ and $\bigvee_{S \in \mathcal{F}} \neg x_S$. These families are far too numerous to list, so we add them only when needed. A SAT solver (CaDiCaL, called through PySAT [IMM18]) proposes a colouring that satisfies the clauses collected so far, and a branch-and-bound search looks in each colour class for a closed family of at least m members. If it finds one, the two clauses for that family are added and the solver is called again. If the search finds nothing, the colouring shows that the value is less than m ; if the solver reports that the formula is unsatisfiable, every colouring has a monochromatic closed family of at least m members. The unit clause $x_\emptyset$ removes the symmetry $\chi \mapsto 1 - \chi$, and letting m increase gives the exact value.

n	1	2	3	4	5	6	7	8	9	10
$f(n)$	1	2	2	3	3	4	4	5	5	6
$F(n)$	1	2	2	3	4	5	7			

Table 1. Exact values. The lower bounds for f are (1.1); the upper bounds for $f(n)$, $n \geq 3$, are certified by the colourings g_n below. The values of $F(5)$, $F(6)$ and $F(7)$ rely on unsatisfiability verdicts, as explained in the text.

Table 1 lists the results, which were checked in three further ways. For $n \leq 4$ all 2^{2^n} colourings were examined directly, without a solver, and the values agree. Every colouring that certifies an upper bound was checked by a second program that shares no code with the search: for each colour class it encodes the existence of a closed subfamily of at least m members as a separate satisfiability problem, with clauses $\neg y_S \vee \neg y_{S'} \vee y_{S \cup S'}$ (and the corresponding clauses for intersections when checking f) and a cardinality constraint $\sum y_S \geq m$ written with the sequential counter of Sinz [Si05]; all these problems are unsatisfiable. Finally, the lower bounds $F(5) \geq 4$, $F(6) \geq 5$ and $F(7) \geq 7$ were recomputed with every family behind an added clause checked to be union-closed, to have at least m members, and to lie in one colour class of the colouring that produced it. These lower bounds therefore depend only on the correctness of the solver’s final unsatisfiability verdict. The computation for $F(7)$ added 256 428 clauses; the one for $f(9)$ added 14 190 clauses and took about twenty minutes on one processor core.

For $n \leq 10$ the value of $f(n)$ is attained by a cardinality colouring. For $3 \leq n \leq 7$ and for $n = 9$ the unrestricted search itself returned such a colouring, but for $n = 8$ it returned one that is not of this kind. The colourings $g_3, \dots, g_7$ below are the ones the search returned; g_8 and g_9 were found by adding the clauses $x_S \leftrightarrow x_{S'}$ for $|S| = |S'|$, and g_{10} , for which the unrestricted search was not run to the end, by listing the cardinality colourings that pass the necessary condition of Section 6 and checking each with the second program. Writing $g_n = (g_n(0), \dots, g_n(n))$, the colourings

$$\begin{aligned} g_3 &= (1, 1, 0, 0), & g_4 &= (1, 1, 0, 1, 0), \\ g_5 &= (1, 1, 0, 1, 0, 0), & g_6 &= (1, 0, 0, 1, 0, 1, 1), \\ g_7 &= (1, 0, 0, 1, 1, 0, 0, 1), & g_8 &= (1, 0, 1, 0, 0, 1, 1, 0, 1), \\ g_9 &= (1, 0, 1, 0, 0, 1, 1, 0, 1, 0), & g_{10} &= (1, 0, 0, 1, 1, 0, 0, 1, 0, 1, 1) \end{aligned}$$

have no monochromatic sublattice with more than $\lceil (n+1)/2 \rceil$ members. For instance, the class of g_9 with colour 1 is $\{0, 2, 5, 6, 8\}$; it contains the progression 2, 5, 8, which yields the sublattice $\{\emptyset, D, D \cup X, D \cup X', D \cup X \cup X'\}$ with $|D| = 2$ and $|X| = |X'| = 3$, and by the computation no monochromatic sublattice has six members.

The table shows F separating from f at $n = 5$. We do not conjecture that $f(n) = \lceil (n+1)/2 \rceil$ for all n .

9. Formal verification

The following statements have been formalised and checked in Lean 4 [MU21] with the Mathlib library [Mat20], version 4.34.1 of both. The development contains no unproved steps, and for each statement below Lean reports that the proof uses only the three standard axioms (propositional extensionality, the axiom of choice and the soundness of quotient types). In the formalisation the ground set is the type of n elements, a colouring is a Boolean function on its subsets, and $F(n)$ and $f(n)$ are defined literally as minima over all colourings of the size of the largest monochromatic family closed under union, respectively under union and intersection. It is also checked that $F(n)$ is the largest m such that every colouring has a monochromatic union-closed family of at least m sets.

- (1) $\lceil (n+1)/2 \rceil \leq f(n) \leq F(n)$ for every n .
- (2) $F(n) \leq \sum_{j < d} \binom{n}{j}$ whenever $nd + 2 < 2^d$; and $F(n) \leq dn^d$ for $n \geq 2$ with $d = 2(\lfloor \log_2 n \rfloor + 1)$.
- (3) For every $\varepsilon > 0$, $F(n) < (1 + \varepsilon)^n$ for all large n ; and $F(n) < 2^n$ for $n \geq 1$.
- (4) For every $t \geq 1$ there are positive integers M and K such that

$$\left(\left\lfloor \frac{\lfloor (n-1)/M \rfloor}{2} \right\rfloor + 1 - t \right)^{t+1} \leq K F(n) \quad (n \geq 1),$$

with subtraction truncated at zero. This is Theorem 1.1 with a constant of the same kind.

- (5) For every $c \in \mathbb{N}$, $n^c < F(n)$ for all large n .
- (6) The first conjecture: there is $\omega : \mathbb{N} \rightarrow \mathbb{R}$ with $\omega(n) \rightarrow \infty$ and $n^{\omega(n)} \leq F(n)$ for every n . The second conjecture: there is $\varepsilon : \mathbb{N} \rightarrow \mathbb{R}$ with $\varepsilon(n) \rightarrow 0$ and $F(n) < (1 + \varepsilon(n))^n$ for every $n \geq 1$. Their conjunction is proved as a single theorem.

The formal proof of (4) follows Sections 2–4 with two changes. The cube lemma is replaced by Mathlib’s Hales–Jewett theorem, as explained in Remark 3.2, and the probabilistic language is replaced by counting: permutations and size vectors are counted rather than sampled, and Lemma 4.2 becomes a double counting over the product of the two sets. The random colouring in (2) is replaced in the same way by a count of colourings. Theorem 1.4 apart from the trivial lower bound, Propositions 5.2 and 7.4, the results of Section 6, Remark 4.3 and the computations of Section 8 are not formalised.

10. Questions

QUESTION 10.1. *What is the order of magnitude of $\log F(n)/\log n$? By Corollary 1.3 and Remark 4.3 it tends to infinity and is at most $(1+o(1))\log_2 n$.*

Is $F(n) = n^{\Theta(\log n)}$? For the random colouring the answer lies between $\log_2 \log_2 n$ and $\log_2 n$ by Proposition 5.2 and the proof of Theorem 1.2, and even there we do not know which end is closer to the truth.

QUESTION 10.2. Is $f(n) = O(n)$? Is $f(n) = \lceil (n+1)/2 \rceil$ for every n ? Proposition 7.4 shows that the random colouring cannot settle the first question.

QUESTION 10.3. Can the factor $(t+2)^{-M(t)}$ in Proposition 3.3 be improved substantially? A density version of Lemma 3.1 with a good bound would improve the growth of ω in Remark 4.3.

The proofs of Theorems 1.1 and 1.2 extend to any fixed number k of colours with only notational changes: in (3.1) the term $2^{2^{M(t-1)}}$ becomes $k^{2^{M(t-1)}}$, the factor $1/2$ in Proposition 2.5 becomes $1/k$, and in the proof of Theorem 1.2 a fixed independent tuple is monochromatic with probability $k^{2^{-2^d}}$. So both conjectures hold for every fixed number of colours. This extension is not formalised.

Data availability

The code, data and Lean formalisation behind this paper are available at <https://doi.org/10.5281/zenodo.XXXXXXX>. Claude (Anthropic) assisted with the coding.

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